

"What is the scientific value of 19th-century phenological records for detecting climate-driven shifts in plant and tree phenology in Central Europe, and how have such records been used to train or validate phenological prediction models?"

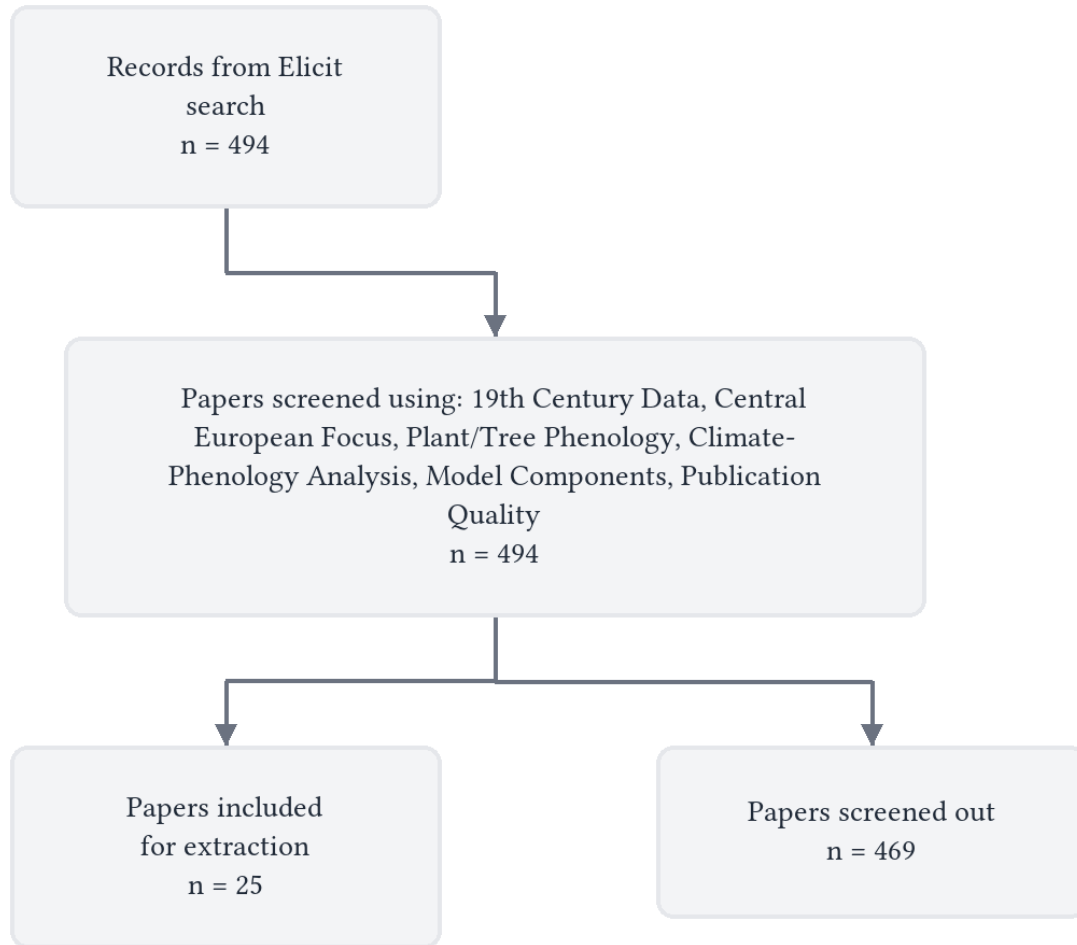
Nineteenth-century phenological records provide unique scientific value by enabling detection of long-term climate-driven shifts, revealing temporal changes in temperature sensitivity invisible to shorter modern datasets, and serving as effective training and validation data for both statistical and mechanistic prediction models that achieve validation confidence ranging from 50-95% depending on application.

Abstract

Nineteenth-century phenological records have proven essential for detecting and quantifying climate-driven shifts in Central European plant and tree phenology, providing temporal baselines impossible to achieve with modern datasets alone. Studies spanning periods from 1702-2020 [1] consistently documented earlier spring phenology, with advances ranging from 1.3-2.1 days/decade [2] and temperature sensitivities of 2-10 days per °C depending on species and region [3]. Historical records enabled detection that 2007 marked the third earliest spring onset in Switzerland across 300 years [4] and that 2018 produced the earliest fruit ripening in Austria since 1768 [5], contextualizing recent changes within natural variability. Critically, these records revealed that temperature sensitivity changed over time, generally increasing but paradoxically decreasing during warming periods (1890-1950; 1970-2007) [4], a finding only possible with multi-century data. Studies also identified species-specific responses, with early-flowering species showing stronger sensitivities (-9.33 days °C⁻¹ for hawthorn versus -5.06 days °C⁻¹ for beech) [6] and better space-time equivalence [7].

Historical records have been extensively used to train and validate phenological prediction models, with both statistical approaches (Bayesian change-point models [8], linear regression [2]) and mechanistic models (ORCHIDEE [9], BALANCE [10]) achieving reasonable performance. Validation confidence varied by application: tree-ring data achieved 95% confidence for extreme climate year reconstruction but only 50% for young tree growth due to selection bias [9], while phenological reconstructions achieved uncertainty of ±3.4 days at interannual timescales [1]. Mechanistic models generally outperformed regression models [6], though both provided consistent insights into temperature forcing and chilling requirements [6]. Models trained on historical data enabled projections showing that late spring species will advance significantly (oak by 24.6 days total by 2040-2069) [6] while early spring species may be delayed due to insufficient chilling [6], demonstrating how century-long records illuminate both past changes and future trajectories.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: *”What is the scientific value of 19th-century phenological records for detecting climate-driven shifts in plant and tree phenology in Central Europe, and how have such records been used to train or validate phenological prediction models?”*

The search returned 494 total results from Elicit.

We retrieved 494 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **19th Century Data:** Does this study utilize phenological observations from the 1800s (1801-1900) as primary or secondary data sources?
- **Central European Focus:** Does this study include data from or analysis of Central European regions (Germany, Austria, Switzerland, Czech Republic, Slovakia, Hungary, Poland, Slovenia, parts of France, Netherlands, Belgium)?
- **Plant/Tree Phenology:** Does this study examine phenological phenomena in terrestrial plants or trees (e.g., leaf emergence, flowering, fruiting, leaf senescence) rather than focusing solely on animal phenology?
- **Climate-Phenology Analysis:** Does this study explicitly examine relationships between climatic variables and phenological timing, or use phenological data to infer climate impacts?
- **Model Components:** Does this study include development, calibration, training, or validation of phenological prediction models using historical data?
- **Publication Quality:** Is this study a peer-reviewed research article that presents quantitative analysis or statistical methods (rather than purely descriptive accounts or grey literature)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Historical Data Characteristics:**

Extract details about 19th-century phenological records used in the study, including:

- Time period covered (start and end years, with emphasis on 19th century data)
- Geographic coverage within Central Europe
- Data collection methods and sources
- Quality assessments or limitations mentioned
- Number of observation sites, years, and species
- Types of phenological phases recorded (leafing, flowering, fruiting, etc.)

- **Climate-Driven Shifts Detected:**

Extract all climate-driven phenological shifts detected using 19th-century records in Central Europe, including:

- Specific phenological changes observed (timing advances/delays)
- Magnitude of changes (days, weeks, rates per decade)
- Time periods when changes occurred
- Species-specific responses to climate change
- Comparison between historical and recent periods
- Evidence linking changes to temperature or other climate variables

- **Scientific Value Assessment:**

Extract authors' assessment of the scientific value of 19th-century phenological records for climate research, including:

- Unique insights only possible with historical data
- Advantages over shorter modern datasets
- Ability to detect long-term trends vs. short-term variability
- Contextualizing recent climate changes
- Limitations or challenges with historical data use
- Comparison of findings with/without historical data

- **Model Applications:**

Extract how 19th-century phenological records were used in prediction models, including:

- Model types developed or tested (statistical, mechanistic, machine learning, etc.)
- Whether historical data was used for training, validation, or both
- Specific model parameters derived from historical records
- Temperature sensitivity coefficients or relationships quantified
- Temporal scope of model applications (hindcasting, forecasting)
- Model complexity and key variables included

- **Model Performance:**

Extract model performance and validation results when historical records were used, including:

- Validation metrics and goodness-of-fit statistics
- Comparison between models with vs. without historical data
- Predictive accuracy for different time periods or species
- Model limitations or failures identified
- Uncertainty estimates or confidence intervals
- Cross-validation results using historical vs. modern data

- **Temperature-Phenology Relationships:**

Extract quantified relationships between temperature and phenology derived from 19th-century records in Central Europe, including:

- Temperature sensitivity coefficients (days advance per °C warming)
- Critical temperature thresholds or growing degree days
- Seasonal temperature effects (spring, winter, previous year)
- Changes in temperature sensitivity over time
- Non-linear responses or threshold effects
- Chilling requirements or other temperature-dependent mechanisms

- **Species and Phases:**

Extract details about plant and tree species and phenological phases studied using 19th-century records, including:

- Scientific and common names of species
- Types of phenological events (bud burst, first leaf, flowering, fruit ripening, etc.)

- Functional groups or ecological categories
- Species-specific findings or differential responses
- Native vs. non-native species
- Focus on Central European flora

- **Future Projections:**

Extract future phenological projections or predictions based on models trained/validated with 19th-century records, including:

- Projected changes under different climate scenarios
- Timeline of projected changes (2050, 2100, etc.)
- Species or phases most vulnerable to future change
- Anticipated changes in species interactions or competition
- Ecosystem-level consequences predicted
- Uncertainty in future projections

Results

Characteristics of Included Studies

The review includes 25 sources examining 19th-century phenological records in Central Europe. Studies varied in geographic scope, temporal coverage, data sources, and analytical approaches.

Study	Full text retrieved?	Time period	Geographic coverage	Data sources	Species/phases	Study focus
F. Badeck et al., 2012 [11]	No	1880-2009	Central Europe (emphasis on Germany)	Plant-Phenological Online Database (PPODB)	Not specified	Database presentation, trend analyses
P. . et al., 2016 [12]	No	1851-1891	Carpathian Basin, Hermannstadt	Few available 19th-century sources	16 plant species	First flowering as temperature proxy
H. Scheifinger et al., 2020 [5]	No	1768-2018	Austria	HISTALP temperature series, phenological observations	Black elder, red currant, apricot	Fruit ripening phenology
Christoph Schleip et al., 2008 [13]	No	Last 250 years	Switzerland, Burgundy (eastern France)	Long-term phenological observations	Grape harvest, spring phenology	Temperature impact assessment
T. Rutishauser et al., 2007 [1]	No	1702-2005	Switzerland	Plant phenological observations	Cherry, apple, beech	Spring season reconstruction

Study	Full text retrieved?	Time period	Geographic coverage	Data sources	Species/phases	Study focus
S. Renner et al., 2021 [2]	Yes	1844-2020	Munich, Germany	Botanical garden observations, herbarium specimens	450 species (focus on <i>Anemone patens</i> , <i>A. pulsatilla</i> , <i>Arum maculatum</i>)	Flowering times, climate data
T. Rutishauser et al., 2008 [4]	No	Last 300 years	Switzerland	Not specified	Three tree species	Spring phenology extremes
Jin-Hyon Jeong et al., 2021 [9]	Yes	Pre-industrial to present	Europe (test sites)	International Tree-Ring Data Bank (ITRDB)	<i>Picea abies</i> , <i>Fagus sylvatica</i>	Tree-ring data for model validation
Anna Knapp et al., 2025 [7]	No	150 years	Vienna, Central Europe	Botanical gardens (11 sites)	17 herbaceous species	Space-for-time substitution analysis
K. Szyga-Pluta et al., 2022 [14]	Yes	1893-2020	Kraków, Potsdam, Poznań, Prague, Toruń, Vienna	National meteorological services	Not specified	Growing season patterns
K. Szyga-Pluta et al., 2022a [15]	No	1893-2020	Prague, Vienna, Toruń, Potsdam	Not specified	Not specified	Growing season patterns
Bianca Plückhahn et al., 2021 [16]	No	1896-present	Geisenheim, Germany	Königlich Preußische Meteorologische Institut	Not specified	Phenological shifts
F. Willems et al., 2022 [17]	No	Last century	Europe	Herbarium specimens (6000+)	20 spring-flowering forest herbs	Flowering phenology
A. Roberts et al., 2015 [6]	Yes	1753-1947	Norfolk, UK; southeastern England	Marsham phenological time series	13 tree species, 1 ground flora species	Phenology prediction models
C. Pfister et al., 2014 [18]	No	1876-2011	Canton Schaffhausen, Switzerland	Official gazette publications	Snowdrop, hazel, horse chestnut, rye, grape vine	Long-term phenological observations

Study	Full text retrieved?	Time period	Geographic coverage	Data sources	Species/phases	Study focus
Jonas Dierenbach et al., 2013 [19]	Yes	1880-2009	Central Europe	Volunteer networks, historical records	Agricultural plants, fruits, wild plants, vines	Online database (PPODB)
C. Defila et al., 2001 [20]	No	1808-1998 (horse-chestnut), 1894-1998 (cherry)	Geneva, Liestal, Switzerland	National observation network	Horse-chestnut, cherry	Phenological trends
A. Molnár et al., 2012 [21]	No	1837-2009	Hungary	Herbarium specimens, field observations	39 orchid taxa	Flowering dates
C. Defila et al., 1992 [22]	No	1808-1991 (chestnut), 1894-1990 (cherry)	Geneva, Liestal, Switzerland	Long-term observations	Chestnut, cherry, summer lime	Environmental indicators
This Rutishauser et al., 2007 [3]	No	Not specified	Switzerland	Not specified	Cherry tree	Spring phenology
Eugenia Mantovani et al., 2026 [23]	Yes	1774-2020	Eastern Alps	Tree-ring data, climate records	Larix decidua, Picea abies, Pinus cembra	Cambial phenology via frost rings
A. Menzel et al., 2005 [8]	No	1900-2003 (Germany), 600 years (Kyoto)	Germany, Kyoto (Japan)	Long-term observational records	Flowering plants, cherry	Bayesian function estimation
A.J.H. van Vliet et al., 2008 [24]	Yes	1868-1968	Netherlands	Volunteer-based observations	250+ species	Phenological changes
I. Kott et al., 2009 [25]	No	Not specified	Middle Bohemia	Not specified	Blackthorn, winter wheat	Long-term observations
H. Pretzsch et al., 2014 [10]	Yes	1870-present	Central Europe (Germany)	Experimental forest plots	Norway spruce, European beech	Forest growth dynamics

Data quality varied across studies. Several investigations explicitly assessed data reliability through biographical verification of observers [18], graphical and statistical verification [18], or sophisticated outlier detection methods [19]. Limitations included temporary gaps during World War II [16], few available data sources for some regions [12], and challenges with big-tree selection bias in tree-ring databases [9].

Climate-Driven Phenological Shifts Detected Using 19th-Century Records

Magnitude and Direction of Changes

Studies consistently documented earlier phenology across Central Europe, though with considerable variation in magnitude. Forest wildflowers flowered over 6 days earlier than at the beginning of the last century, with flowering advancing 3.6 days per 1°C warming [17]. Three species with short flowering times in Munich showed mean advances of 1.3-2.1 days/decade or 3.2-4.2 days/1°C warming over 176 years [2]. Specifically, *Anemone pulsatilla* advanced 2.1 days/decade [2], *A. patens* 1.6 days/decade [2], and *Arum maculatum* 1.3 days/decade [2].

Orchids in Hungary showed that 31 of 39 taxa (79%) displayed apparent advancement in mean flowering time, with nine taxa showing statistically significant changes [21]. The average advancement was 3 days over the last 50 years compared to pre-1960 periods [21], with significant advancements reaching 7.7 days (8.6% of the flowering period) [21] and the most extreme advancement reaching 13.9 days [21].

Swiss spring phenology showed a mean trend of 1.5 days/decade from 1965-2002 [3], with earlier onset of spring phenological phases by 2 to 10 days per 1°C temperature increase depending on species and region [3]. Herbaceous species in Vienna botanical gardens advanced flowering by seven days for a 1°C temperature increase [7].

The Netherlands exhibited particularly pronounced changes, with flowering, leaf unfolding, and fruit ripening beginning 10 to 15 days earlier after the late 1980s compared to before 1990 [24], extending the growing season by about a month [24].

Growing Season Changes

Growing season length increased across Central Europe, though drivers varied by location. In Prague, the growing season start advanced at -2.09 days/decade with end dates delayed at +1.28 days/decade, resulting in prolongation of +3.88 days/decade [14]. In Toruń and Potsdam, increases were primarily caused by later end dates, while in other stations earlier start dates determined the prolongation [15]. Intensification of changes occurred recently, distinguishing two subperiods with different intensities [14, 15].

Temporal Patterns and Extreme Events

The year 2007 was characterized by record high temperatures leading to very early onset of plant phenological spring phases, with 98 out of 302 possible observations (32%) setting record early dates in Switzerland [4]. In the context of the last 300 years, 2007 marked the third earliest plant phenological spring onset in Switzerland after 1961 and 1794 [4].

The spring of 2018 in Austria produced the earliest beginning of fruit ripening dates since 1946 for black elder and red currant, and the second earliest for apricot [5]. Even when considering lower bounds of confidence intervals from reconstructed data, 2018 remained the earliest entry date for black elder since 1768, with apricot ranking fourth after 1811, 1794, and 1797 [5].

Historical comparisons revealed that recent decades do not show a preponderance of very early years as might be expected from increased spring temperatures alone [1]. The 1940s, 1910s, 1890s, and early 18th century showed similarly early starts of spring season compared to recent decades [1]. However, moving linear trend analysis demonstrated unprecedented agreement towards earlier spring onsets in observed and temperature-based reconstructed plant phenological records in the late 20th/early 21st century [1].

Species-Specific Responses

Different species and phenological phases showed varying sensitivities to temperature. Grape harvest dates showed the highest Bayesian correlations with June temperatures of the current year [13]. Spring phenological phases were influenced by May temperatures of the current year and summer temperatures of the preceding growing season [13].

Among orchids, the strongest responses occurred in species that flower relatively early in spring, exhibit autogamous or deceptive pollination mechanisms, have long life spans, and possess Mediterranean centers of distribution [21]. For herbaceous species, space-time equivalence was better fulfilled for early-flowering species than for late-flowering ones, suggesting differential climate sensitivity across temporal niches [7].

Long-term phenological observations showed that snowdrop and hazel exhibited earlier flowering related to global warming, while grape vine showed later occurrence of flowering [18]. Different phenophases and plant species reacted differently to environmental influences, with earlier appearance dates in spring and weak tendencies towards later appearance dates in autumn [20].

Cambial phenology in Alpine conifers showed shifts of approximately 7 days earlier per century or per °C of warming over the past 200 years [23], with the timing shifting from the first half of June in the early 1800s to the last week of May in recent decades [23]. Notably, no significant differences in this shift occurred across three conifer taxa (*Larix decidua*, *Picea abies*, *Pinus cembra*) [23].

Temperature Sensitivity Changes Over Time

Temperature sensitivity of Swiss plant phenology changed within the last three centuries, generally increasing over the record period but also decreasing during two periods with warming trends (1890-1950; 1970-2007) [4]. At Geisenheim, Germany, flowering onset changes showed advancing rates reaching up to -1.7 days/year in 2003 [8], with maximum change point probability occurring in the mid-1980s [8].

Forest growth dynamics in Central Europe revealed that dominant tree species Norway spruce and European beech exhibited significantly faster tree growth (+32 to 77%), stand volume growth (+10 to 30%), and standing stock accumulation (+6 to 7%) compared to 1960 [10]. The growing season extended by 10.8 days since the early 1960s and 22 days over the last 110 years, with European beech profiting more than Norway spruce [10].

Scientific Value of Historical Phenological Records

Enabling Long-Term Trend Detection

Historical records provided unique insights impossible with shorter modern datasets. The continuous, annually resolved reconstruction of Swiss spring from 1702 to 2005 offered high potential for applications in long-term climate impact studies and vegetation model validations [1]. The 280-year series of cherry tree flowering provided a unique possibility to assess climate impact on spring phenological variability during past centuries [3].

Historical data enabled detection of long-term trends versus short-term variability that shorter datasets could not resolve [5, 8, 14, 17, 20, 21, 23]. Long-term experimental plots provided 140-year time series data on forest stand dynamics essential for understanding long-term environmental impacts [10].

Contextualizing Recent Climate Changes

Historical records placed recent events in broader temporal context. The 2018 spring in Austria was contextualized within a 251-year perspective [5]. Long-term data allowed comparison of recent climate changes with earlier periods

[1, 4, 6, 8], providing baselines for assessing whether contemporary changes were unprecedented [10].

Tree-ring data provided annual information on historical tree growth and physiology before significant human-induced environmental changes, offering century-long perspectives essential for understanding transitions from pre-industrial to present-day conditions [9].

Improving Model Development and Validation

Historical data enhanced model validation and provided long-term validation tests for land-surface models [9]. Combined time series corrected for artefacts in simple averages, providing more accurate pictures of climate-driven shifts [19]. Historical records helped distinguish between long-term trends and short-term variability, crucial for understanding climate change impacts [14].

Herbarium collections provided long-term data for evaluating importance of life-history components in driving species-specific responses to climate change [21]. The long-term nature of records allowed detection of long-term trends and provided baselines for understanding climate-driven shifts [6].

Limitations and Challenges

Despite their value, historical records presented challenges. Statistical reconstruction uncertainty of ± 3.4 days at interannual timescale was noted for Swiss spring plant reconstructions [1]. Combining historical observations with herbarium specimens introduced noise [2]. Considerable uncertainty ranges existed in rates of change calculations [8].

Big-tree selection bias in tree-ring databases limited confidence for certain benchmarks, with only 50% of test cases showing consistency for young tree growth when using International Tree-Ring Data Bank data versus unbiased datasets [9]. Limited availability of long-term data restricted opportunities for temporal analysis [7], and uncertainty remained about whether patterns found in botanical gardens applied to less-controlled habitats [7].

Care was required when processing phenological data, considering that growth and development were determined by numerous and complex environmental factors [22]. The need for careful model selection and awareness of potential measurement errors was emphasized [6].

Model Applications of Historical Records

Model Types and Approaches

Studies employed diverse modeling approaches. Statistical models included Pearson correlation coefficients and linear moving window trends [13], Bayesian correlation and model comparison approaches [13], linear-mixed effect models [7], multiple regression [5], and stepwise linear regression [24]. Mechanistic models included ecophysiological models such as BALANCE [10] and ORCHIDEE [9].

The Bayesian approach applied to Swiss and Burgundy data calculated model probabilities, change-point probabilities, functional descriptions, and rates of change of three models with increasing complexity and temperature weights of single months [13]. For German flowering records, the one-change-point model was preferred over less sophisticated alternatives (linear model and constant model) [8].

A. Roberts et al. (2015) compared both statistical regression and mechanistic thermal sensitivity phenology models, finding that mechanistic models performed best in the majority of cases, though both provided remarkably consistent insights into relative sensitivity of each species to forcing and chilling effects [6].

Data Usage: Training and Validation

Historical records served both training and validation purposes. Austrian fruit ripening phenology used multiple regression driven by pre-season mean monthly temperatures to extend time series back to 1768 [5]. Swiss and Burgundy data were used for both training and validation of temperature impact models [13].

Munich flowering data from 1844 were used for training linear models relating flowering dates to air temperature with pre-season lengths of 30, 60, or 90 days [2]. The Marsham phenological time series trained and validated phenological prediction models, with historical data used to parameterize models tested against projected climate data [6].

Tree-ring data from the International Tree-Ring Data Bank served validation purposes for land-surface models [9], with verification processes comparing model outputs against both big-tree and all-tree datasets [9]. Growing season data from 1893-2020 were used for both training and validation of mathematical formulas for growing season start and end dates based on 5°C air temperature thresholds [14].

Netherlands phenological data from 1868-1968 were used for training and validation of stepwise linear regression models examining correlations between temperature and phenological events [24]. Historical data from experimental forest plots trained and validated the ecophysiological model BALANCE for hindcasting (1901-1930) and forecasting (1981-2010) [10].

Temperature Sensitivity Coefficients Derived

Models quantified temperature-phenology relationships with precision. Munich species showed temperature sensitivity coefficients of -3.1 to -4.6 days/°C for *Anemone pulsatilla*, -2.6 to -3.2 days/°C for *A. patens*, and -3.6 to -4.2 days/°C for *Arum maculatum* [2]. European forest wildflowers advanced 3.6 days per 1°C warming [17].

The Marsham time series revealed temperature sensitivities ranging from -5.06 days °C⁻¹ for beech to -9.33 days °C⁻¹ for hawthorn [6], with chilling slope estimates of 7.36 days °C⁻¹ for hawthorn and 5.39 days °C⁻¹ for birch [6]. Swiss spring data showed 2-10 days advance per 1°C warming depending on species and region [3].

Growing season analysis quantified that a 1°C increase in spring temperature corresponded with growing season start occurring 5-6 days earlier [14]. Cambial phenology in Alpine conifers advanced approximately 7 days per century or per °C of warming [23].

Model Parameters and Complexity

Models incorporated various parameters derived from historical records. The Austrian fruit ripening model used pre-season mean monthly temperatures as key variables [5]. Swiss and Burgundy models included temperature weights of single months, with three selected models of increasing complexity [13].

Swiss spring reconstruction parameters included weighted means for flowering of cherry and apple trees and bud-burst of beech [1]. Munich models used flowering dates and air temperature with variable pre-season lengths [2].

Growing season models employed air temperature thresholds of 5°C and mathematical formulas for start and end dates [14]. Marsham models included daily temperatures, chilling requirements in the range of 10-17°C, and forcing functions [6]. Netherlands models incorporated climate variables including temperature and precipitation [24].

Cambial phenology models used tree age, frost ring frequency, and year as variables, with statistical models including the anomaly method and Linear Mixed Effects Models [23]. The effect of year on frost ring formation was estimated at -0.070 ± 0.023 days year⁻¹ [23].

Forest growth models (BALANCE) included changes in atmospheric CO₂ concentrations, nitrogen depositions, temperature, and precipitation as key variables [10].

Temporal Scope of Applications

Most model applications focused on hindcasting to understand past changes. Swiss and Burgundy models applied Bayesian analyses to detect major changes in long-term phenological and temperature time series [13]. Austrian fruit ripening models extended phenological time series from 1946 observations back to 1768 [5].

Swiss spring reconstructions spanned 1702-2005 [1]. Munich flowering models covered 1844-2020 [2]. Growing season analyses examined 1893-2020 [14]. German flowering records used Bayesian change-point models for hindcasting throughout the 20th century [8].

Some models extended to forecasting applications. The Marsham phenological time series supported both hindcasting and forecasting under a fossil fuel intensive SRES scenario for periods 2010-2039 and 2040-2069 [6]. Netherlands models forecast future changes in phenological events based on projected climate scenarios [24]. Forest growth models hindcast for 1901-1930 and forecast for 1981-2010 [10].

Growing season projections indicated that start dates would occur 4.7-10.1 days earlier in 2040-2059 and 4.3-21.6 days earlier in 2080-2099 [14].

Model Performance and Validation Results

Validation Metrics

Model performance varied across approaches and species. For temperature estimation using first flowering data as proxies, accuracy ranged from 1.0°C to 1.5°C [12]. Austrian fruit ripening models accommodated uncertainty through lower (5%) and upper (95%) bounds of confidence intervals, accurately predicting dates with 2018 remaining the earliest for black elder even when considering lower bounds [5].

Swiss spring plant reconstructions achieved statistical reconstruction uncertainty of ± 3.4 days at interannual timescale, with unprecedented agreement towards earlier spring onsets in observed and temperature-based records [1]. Munich flowering models produced temperature sensitivity ranges for each species, though combining historical and modern data introduced some noise [2].

Tree-ring model validation achieved 95% confidence for extreme climate years, 70% confidence for mature trees and size-related growth, but only 50% confidence for young trees [9]. Higher confidence was observed for beech (84%) compared to spruce (64%) [9]. High correlation coefficients ($r > 0.90$) between observed and modeled temperature series validated climate reconstructions [23], with all correlations significant at $p < 0.001$ [23].

Marsham phenology models achieved R^2 values ranging from 29-73% for time-window and photoperiod-sequential reaction models [6]. Netherlands phenological models explained 62% of annual variability using climate variables through stepwise linear regression [24], with temperature explaining 76% of variability in spring events and 40% in autumn events [24].

Model Comparisons

When comparing model types, mechanistic models generally outperformed regression models, except for wood anemone, sycamore, horse chestnut, and maple [6]. Both mechanistic and regression models provided remarkably consistent insights into relative sensitivity of species to forcing and chilling effects [6].

For Bayesian change-point models, the one-change-point model was preferred over linear and constant models when analyzing long-term flowering records [8], though considerable uncertainty ranges existed in rate estimates [8].

Tree-ring validation showed that using International Tree-Ring Data Bank data would result in the same conclusions as using unbiased European biomass network data in approximately 95% of test cases for extreme climate years, 70% for mature tree growth and size-related trends, but only 50% for young tree growth [9].

Model Limitations

Several limitations were identified. Some species showed presence of temporal autocorrelation in residuals [6]. Leave-one-out cross-validation validated interpolation techniques for temperature reconstruction [23], though no significant improvement occurred with AR(1) correlation structure [23].

For forest growth dynamics, climate change alone did not fully explain observed growth trends; changes in air chemistry (CO₂ and nitrogen deposition) were necessary for better alignment between model outputs and empirical data [10]. This highlighted the complexity of factors beyond temperature affecting phenological and growth responses.

Big-tree selection bias in tree-ring databases created limitations, particularly for benchmarking young tree growth [9, 9]. Combining historical observations with herbarium specimens introduced noise into analyses [2]. The need for spatial modeling when analyzing herbarium data across large scales was emphasized to avoid pseudoreplication and geographical biases [17].

Temperature-Phenology Relationships Quantified

Preseason Temperature Effects

Analyses revealed complex seasonal temperature influences on phenology. Grape harvest dates showed highest Bayesian correlations with June temperatures of the current year [13]. Spring phenological phases were influenced by May temperatures of the current year and summer temperatures of the preceding growing season [13].

Munich flowering species responded to preseason temperatures with varying lengths. Temperature sensitivity coefficients were quantified as -3.1 to -4.6 days/°C for *Anemone pulsatilla*, -2.6 to -3.2 days/°C for *A. patens*, and -3.6 to -4.2 days/°C for *Arum maculatum* [2], with optimal preseason lengths of 30, 60, or 90 days depending on species [2].

Growing season start and end fluctuations were closely correlated with mean monthly temperatures in April and October, respectively [14]. Temperature explained 76% of variability in spring events and 40% in autumn events in the Netherlands [24].

Chilling and Forcing Requirements

The Marsham phenological time series revealed that all species were sensitive to spring forcing, but vernal and northern European species were additionally responsive to cold temperatures in the previous autumn [6]. Critical temperature thresholds for chilling ranged from 10 to 17°C [6], with chilling slope estimates of 7.36 days °C⁻¹ for hawthorn and 5.39 days °C⁻¹ for birch [6].

Species with early phenology accumulated forcing units at lower temperatures [6]. Temperature sensitivity coefficients for forcing ranged from -5.06 days °C⁻¹ for beech to -9.33 days °C⁻¹ for hawthorn [6].

Threshold Effects and Non-Linearities

Growing season models commonly employed a 5°C threshold temperature for both start and end dates [14], with a 1°C increase in spring temperature generally corresponding to a 5-6 day earlier start [14].

Frost rings in Alpine conifers formed when sub-zero temperatures persisted for at least 2 days, reaching average minimums of -2°C [23]. Cold spells responsible for frost ring formation remained consistent over time, typically involving temperatures dropping below freezing for an average of 4-5 consecutive days [23].

Changes in temperature sensitivity over time revealed non-linear patterns. Swiss plant phenology showed generally increasing sensitivity over the last three centuries, but sensitivity decreased during two periods with warming trends (1890-1950; 1970-2007) [4]. This suggested complex interactions between temperature changes and phenological responses that varied across different climate contexts.

Future Climate Scenarios and Projections

The Marsham phenological time series supported projections under a fossil fuel intensive emissions scenario (A1F1) for periods 2010-2039 and 2040-2069 [6]. Models predicted that late spring species (sweet chestnut, oak, beech, ash) would advance significantly, while early spring species (hawthorn, birch) might be delayed or advance less due to insufficient chilling [6].

Projected changes included oak advancing by 14.3 days by 2010-2039 and another 10.3 days by 2040-2069 [6]. Changes in relative timing between birch and oak were anticipated [6], with potential consequences for competitive advantage for light, understorey growth, and seedling recruitment [6]. Shifts in species abundance and community composition were predicted as more likely than genetic adaptation [6], though the net effect on forest communities remained unknown [6].

Growing season projections based on Central European data indicated that start dates would occur 4.7-10.1 days earlier in 2040-2059 and 4.3-21.6 days earlier in 2080-2099 [14], with observed trends expected to continue or increase [14].

Netherlands projections anticipated significant temperature increases by 2100, with winter temperatures rising 1.8-4.7°C and summer temperatures rising 1.8-5.7°C [24]. Plant and animal species faced risks of local extinction with warming above 2-3°C [24], alongside substantial changes in ecosystem structure and functioning [24]. Increased frequency of extreme weather events was expected [24], with uncertainty depending on success in controlling energy use and reducing greenhouse gas emissions [24].

Forest growth projections suggested that European beech may become a stronger competitor, potentially outcompeting Norway spruce [10]. Increased carbon turnover rates were anticipated, with potential for earlier harvests and increased annual cuts, but also risks including storm damage and nutrient export [10]. High uncertainty remained regarding whether growth trends would remain positive under future climate change [10].

Synthesis

The accumulated evidence from 19th-century phenological records demonstrates clear climate-driven advances in spring phenology across Central Europe, yet with important nuances in magnitude, timing, and species responses that require careful interpretation.

Reconciling Variation in Advancement Rates

Studies reported phenological advances ranging from 1.3 days/decade [2] to 2.1 days/decade [2] for individual species, with temperature sensitivities spanning 2-10 days per °C depending on species and region [3]. This variation reflects genuine biological differences rather than methodological inconsistencies. Species flowering early in spring showed stronger temperature sensitivities (-9.33 days °C⁻¹ for hawthorn) [6] compared to later-flowering species (-5.06 days °C⁻¹ for beech) [6]. The better space-time equivalence for early-flowering species [7] confirms this pattern extends beyond individual datasets.

Geographic variation also contributed. Studies from southern locations (Prague, Vienna) showed higher year-to-year variability in growing season start [15] compared to northern sites, while the mechanisms of prolongation differed: Toruń and Potsdam showed increases driven by later end dates, whereas Prague and Vienna showed increases driven by earlier start dates [15]. These geographic patterns likely reflect latitude-dependent differences in limiting factors, with northern sites more constrained by autumn temperatures and southern sites more responsive to spring warming.

Temporal Complexity in Temperature Sensitivity

A critical finding across multiple studies was that temperature sensitivity of phenology changed over time, generally increasing but paradoxically decreasing during warming periods (1890-1950; 1970-2007) [4]. This apparent contradiction can be explained mechanistically: as spring temperatures rise above historical ranges, species may encounter physiological constraints or insufficient chilling requirements that reduce their responsiveness to further warming. This is consistent with findings that vernal and northern European species respond to cold temperatures in the previous autumn [6], with critical chilling thresholds of 10-17°C [6].

The differential response to warming periods helps explain why recent decades do not show a preponderance of very early years despite increased spring temperatures [1], with the 1940s, 1910s, and 1890s showing similarly early spring onsets [1]. However, moving linear trend analysis revealed unprecedented agreement towards earlier onsets in the late 20th/early 21st century [1], suggesting that recent warming has pushed some species beyond historical variability ranges even if individual years are not extreme relative to earlier periods.

Distinguishing Climate Effects from Other Drivers

Not all observed changes were purely climate-driven. The earlier budburst of horse-chestnut trees in Geneva was attributed mainly to urban heat island effects rather than regional climate change, a phenomenon not observed in cherry trees at Liestal [20]. Anthropogenic shifts in harvest timing (hay, grape) must be distinguished from biological responses [18].

For forest growth dynamics, climate change alone did not fully explain observed increases in tree growth (+32-77%) and stand volume growth (+10-30%) [10]. The BALANCE model required incorporation of atmospheric chemistry changes (CO₂ fertilization and nitrogen deposition) to align with empirical data [10, 10], indicating that phenological shifts occur within broader physiological responses to multiple environmental changes.

Model Performance Hierarchy and Validation Confidence

Model validation results reveal a clear hierarchy in data reliability. Tree-ring data achieved 95% confidence for reconstructing extreme climate years but only 50% confidence for young tree growth due to big-tree selection bias [9]. This demonstrates that the same historical dataset can have high utility for some applications (extreme event detection) while requiring caution for others (age-class specific growth).

Mechanistic models generally outperformed statistical models [6], yet both approaches provided consistent insights into forcing and chilling sensitivities [6]. The preference for one-change-point models over linear models in Bayesian analyses [8] reflects the non-linear nature of phenological change, with shifts concentrated around the mid-1980s [8] rather than gradual across decades.

For applications requiring high spatial resolution, combining herbarium data with spatial modeling was essential to avoid biases from data clustering [17]. Without spatial autocorrelation considerations, estimates of phenological shifts were biased and model fits poor [17], demonstrating that methodological rigor in handling historical data directly affects conclusion validity.

Implications for Future Projections

Future projections reveal potential for phenological disruption beyond simple advancement. Predicted changes in the order of spring events [6] could alter competitive dynamics, with oak advancing 14.3 days by 2010-2039 and another 10.3 days by 2040-2069 [6], while early spring species may advance less or even be delayed due to insufficient chilling [6]. This desynchronization threatens species interactions, particularly for vernal ground flora competing for light beneath canopy trees.

The finding that cambial phenology in Alpine conifers showed smaller shifts (7 days per century) [23] than phenology in other plant compartments in the same region indicates that different physiological processes respond to warming at different rates. This has implications for carbon dynamics, as the critical role of cambial activity in forest carbon storage may not track temperature changes as closely as foliar phenology.

Growing season projections of 4.3-21.6 days earlier start by 2080-2099 [14] suggest continued intensification, though uncertainty remains high regarding whether observed growth trends will persist [10]. The substantial risk to species facing local extinction with warming above 2-3°C [24] underscores the urgency of these changes.

References

1. T. Rutishauser, J. Luterbacher, F. Jeanneret, et al (2007) A phenology-based reconstruction of interannual changes in past spring seasons. <https://doi.org/10.1029/2006JG000382>
2. S. Renner, Markus Wesche, C. Zohner (2021) Climate data and flowering times for 450 species from 1844 deepen the record of phenological change in southern Germany. *American-Eurasian journal of botany*. <https://doi.org/10.1002/ajb2.1643>
3. This Rutishauser, S. Studer (2007) Klimawandel und der Einfluss auf die Frühlingsphänologie. <https://doi.org/10.3188/SZF.2007.0105>
4. T. Rutishauser, J. Luterbacher, C. Defila, et al (2008) Swiss spring plant phenology 2007: Extremes, a multi-century perspective, and changes in temperature sensitivity. <https://doi.org/10.1029/2007GL032545>
5. H. Scheifinger (2020) The extreme warmth of the Central European spring in 2018 and its effects on fruit ripening phenology in Austria – a 251 year perspective. <https://doi.org/10.5194/egusphere-egu2020-2968>
6. A. Roberts, Christine Tansey, R. Smithers, Albert B. Phillimore (2015) Predicting a change in the order of spring phenology in temperate forests. *Global Change Biology*. <https://doi.org/10.1111/gcb.12896>

7. Anna Knapp, Wolfgang Matzke, Barbara Knickmann, et al (2025) Space-for-Time Substitution to study changes in flowering phenology in herbaceous species is most reliable for early-flowering species. *Basic and Applied Ecology*. <https://doi.org/10.1016/j.baae.2025.10.007>
8. A. Menzel, V. Dose (2005) Analysis of long-term time series of the beginning of flowering by Bayesian function estimation. <https://doi.org/10.1127/0941-2948/2005/0040>
9. Jin-Hyon Jeong, J. Barichivich, P. Peylin, et al (2021) Using the International Tree-Ring Data Bank (ITRDB) records as century-long benchmarks for global land-surface models. *Geoscientific Model Development*. <https://doi.org/10.5194/GMD-14-5891-2021>
10. H. Pretzsch, P. Biber, G. Schütze, et al (2014) Forest stand growth dynamics in Central Europe have accelerated since 1870. *Nature Communications*. <https://doi.org/10.1038/ncomms5967>
11. F. Badeck, J. Schaber (2012) MORE THAN A CENTURY OF PHENOLOGICAL OBSERVATIONS: MODEL ANALYSES AND ISSUES RELATED TO THE ASSESSMENT OF CLIMATE CHANGE EFFECTS
12. P. ., Ongrácz, S. ., Zentkirályi (2016) TESTING PLANT PHENOPHASE AS PROXY : SENSITIVITY ANALYSIS OF FIRST FLOWERING DATA FROM THE 19 TH CENTURY
13. Christoph Schleip, T. Rutishauser, J. Luterbacher, A. Menzel (2008) Time series modeling and central European temperature impact assessment of phenological records over the last 250 years. <https://doi.org/10.1029/2007JG000646>
14. K. Szyga-Pluta, A. Tomczyk, K. Piotrowicz, E. Bednorz (2022) Patterns in the Multiannual Course of Growing Season in Central Europe Since the End of the 19th Century. *Quaestiones Geographicae*. <https://doi.org/10.14746/quageo-2023-0005>
15. K. Szyga-Pluta, A. Tomczyk, K. Piotrowicz, E. Bednorz (2022) Patterns in the Multiannual Course of Growing Season in Central Europe Since the End of the 19th Century. <https://doi.org/10.2478/quageo-2023-0005>
16. Bianca Plückhahn, A. Engels, Andreas Ehlig (2021) Verursacht der Klimawandel eine sprunghafte Veränderung in der Pflanzenentwicklung ? <https://doi.org/10.5194/dach2022-148>
17. F. Willems, J. F. Scheepens, O. Bossdorf (2022) Forest wildflowers bloom earlier as Europe warms: lessons from herbaria and spatial modelling. *New Phytologist*. <https://doi.org/10.1111/nph.18124>
18. C. Pfister, O. Wetter (2014) Eine tragfähige Datenbrücke – Phänologische Beobachtungen im Amtsblatt des Kantons Schaffhausen 1876–1950. *Bauhinia*. <https://doi.org/10.7892/BORIS.64766>
19. Jonas Dierenbach, F. Badeck, J. Schaber (2013) The plant phenological online database (PPODB): an online database for long-term phenological data. *International journal of biometeorology*. <https://doi.org/10.1007/s00484-013-0650-2>

20. C. Defila, B. Clot (2001) Phytophenological trends in Switzerland. *International journal of biometeorology*. <https://doi.org/10.1007/S004840100101>
21. A. Molnár, J. Tökölyi, Z. Végvári, et al (2012) Pollination mode predicts phenological response to climate change in terrestrial orchids: a case study from central Europe. <https://doi.org/10.1111/j.1365-2745.2012.02003.x>
22. C. Defila (1992) [Phenology, an indicator for environmental changes]. *Schweizerische Rundschau für Medizin Praxis = Revue suisse de médecine Praxis*
23. Eugenia Mantovani, A. L. Prendin, Michele Brunetti, et al (2026) Phenological Shifts in Wood Formation Tracked by Frost Rings Across Two Centuries. *Global Change Biology*. <https://doi.org/10.1111/gcb.70745>
24. A.J.H. van Vliet (2008) Monitoring, analysing, forecasting and communicating phenological changes. <https://doi.org/10.18174/16350>
25. I. Kott, K. Helman (2009) TWO CENTURIES OF PHENOLOGICAL OBSERVATION IN MIDDLE BOHEMIA
DUE SECOLI DI OSSERVAZIONI FENOLOGICHE NELLA MEDIA BOEMIA